How-To Library

Informatica

Performance Tuning Guidelines to Read Data from or Write Data to Amazon S3

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Abstract

When you read data from or write data to Amazon S3, multiple factors such as hardware parameters, Hadoop cluster parameters, and mapping parameters impact the operation performance. You can optimize the performance by tuning these parameters appropriately. This article describes general guidelines to help you tune the operation performance when you read data from or write data to Amazon S3.

Supported Versions

- PowerExchange® for Amazon S3 10.2 or later

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Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the key hardware, Hadoop cluster, and Informatica mapping parameters that you can tune to optimize the performance of read and write operations for Amazon S3.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance might vary based on individual environments and other parameters even when you use the same data.

Performance Tuning Areas

You can optimize the performance by tuning the following areas:

- Hardware
- Hadoop cluster properties
- Informatica mapping
- Hive connection on Amazon S3

Tune the Hardware

You can tune the following hardware parameters to optimize the performance of the machine where the Data Integration Service runs:

- CPU frequency
- NIC card ring buffer size

CPU Frequency

Dynamic frequency scaling dynamically adjusts the processor frequency either for power savings or to reduce heat. Ensure that the CPU operates at least at the base frequency.

When CPUs are underclocked, where they run below the base frequency, the performance degrades by 30% to 40%. Informatica recommends that you work with your IT system administrator to ensure that all the nodes on the cluster are configured to run at their supported base frequency.

To tune the CPU frequency for Intel multicore processors, perform the following steps:

1. Run the lscpu command to determine the current CPU frequency, base CPU frequency, and the maximum CPU frequency that the processor supports.
2. Request your system administrator to perform the following tasks:
   a. Increase the CPU frequency to the supported base frequency.
   b. Change the power management setting to **OS Control** at the BIOS level.
3. Run CPU-intensive tests to monitor the CPU frequency in real time and adjust the frequency for improved performance. On Red Hat operating systems, you can install a monitoring tool such as cpupower.
4. Work with your IT department to ensure that the CPU frequency and power management settings are persisted even for future system reboots.

NIC Card Ring Buffer Size

NIC configuration is a key factor in network performance tuning. When you deal with large volumes of data, it is crucial that you tune the Receive (RX) and Transmit (TX) ring buffer size. The ring buffers contain descriptors or pointers to the socket kernel buffers that hold the packet data.

You can run the ethtool command to determine the current configuration. For example, run the following command:

```
# ethtool -g eth0
```

The following sections show a sample output:

```
Ring parameters for eth0:
Pre-set maximums:
RX: 2040
RX Mini: 0
RX Jumbo: 8160
TX: 255

Current hardware settings:
RX: 255
RX Mini: 0
RX Jumbo: 0
TX: 255
```

The **Pre-set maximums** section shows the maximum values that you can set for each parameter. The **Current hardware settings** section shows the current configuration details.

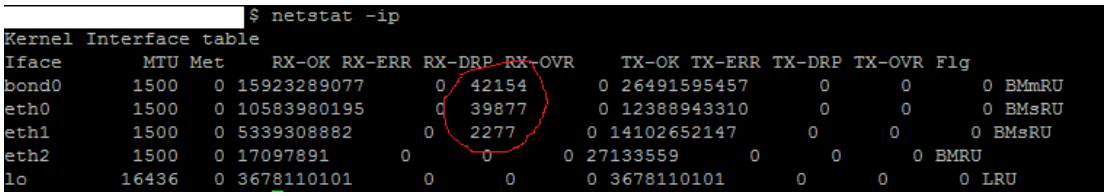
A low buffer size leads to low latency. However, low latency comes at the cost of throughput. For greater throughputs, you must configure large buffer ring sizes for RX and TX.

Informatica recommends that you use the ethtool command to determine the current hardware settings and the maximum supported values. Then, set the values based on the maximum values that are supported for each operating system. For example, if the maximum supported value for RX is 2040, you can use the ethtool command as follows to set the RX value to 2040:

```
# ethtool -G eth0 RX 2040
```

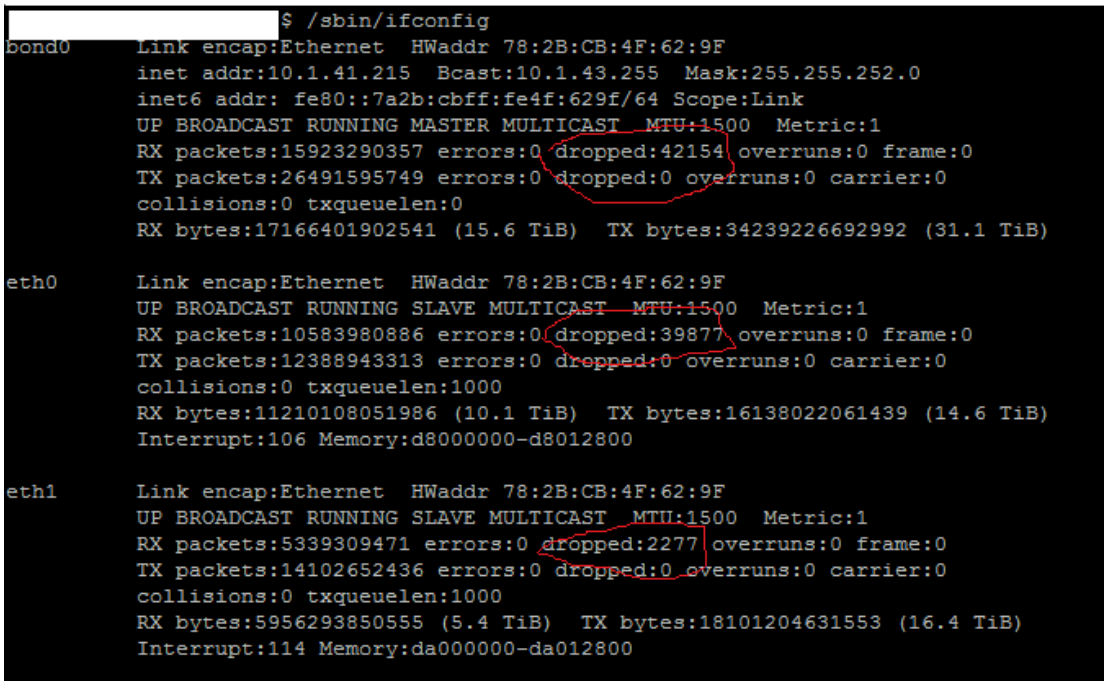
If you set a low ring buffer size for data transfer, packets might get dropped. To find out if packets were dropped, you can use the netstat and ifconfig commands.

The following image shows a sample output of the netstat command:



The RX-DRP column indicates the number of packets that were dropped. Set the RX value such that no packets get dropped and the RX-DRP column shows the values as 0. You might need to test several values to optimize the performance.

The following image shows a sample output of the ifconfig command:



The status messages indicate the number of packets that were dropped.

Tune the Hadoop Cluster and Data Integration Service

You can tune the Hadoop configuration properties for the Data Integration Service in the following files:

- `hive-site.xml`
- `hadoopEnv.properties`

hive - hive-site.xml

Configure the following properties in the `hive-site.xml` file located at:

`$INFA_HOME/services/shared/Hadoop/<distr>/conf/hive-site.xml`

mapred.compress.map.output

Determines whether the output of the map phase is compressed. Default is false. Informatica recommends you to tune the parameter for better performance by setting the parameter to true.

mapred.map.output.compression.codec

Specifies the compression codec used for map output compression. Default is `org.apache.hadoop.io.compress.DefaultCodec`. Informatica recommends using Snappy codec `org.apache.hadoop.io.compress.SnappyCodec` for better performance.

mapred.map.tasks.speculative.execution

Specifies whether you can speculatively run the map tasks. Default is true. With speculative map task execution, duplicate tasks are spawned for the tasks that do not make progress. Original and speculative tasks are considered alike. The task that completes first is considered and the other is killed.

Informatica recommends keeping the default value set to true for better performance.

mapred.reduce.tasks.speculative.execution

Specifies whether you can speculatively run the reduce tasks. Default is true. The `mapred.reduce.tasks.speculative.execution` property is similar to map task speculative execution in functionality. Informatica recommends setting `mapred.reduce.tasks.speculative.execution` to false to disable the property.

mapred.min.split.size and mapred.max.split.size

Use `mapred.min.split.size` and `mapred.max.split.size` properties in conjunction with the `dfs.block.size` property. These parameters impact the number of input splits and hence the parallelism.

Informatica recommends using the following formula for each map task on a data block:

`mapred.min.split.size < dfs.block.size < mapred.max.split.size`

Calculate the input split size with the following formula: `max(minimumSize, min(maximumSize, blockSize))`

hive.exec.compress.intermediate

Determines whether the results of intermediate map and reduce jobs in a Hive query are compressed. Default is false. The `hive.exec.compress.intermediate` property is not the same as `mapred.compress.map.output` that deals with the compression of the output of the map task.

Informatica recommends setting the `hive.exec.compress.intermediate` property to true. `hive.exec.compress.intermediate` uses the same codec specified by `mapred.output.compression.codec`. Informatica recommends SnappyCodec for better performance.

To determine the number of reduce tasks for a mapping, you can set the following properties in the `hive-site.xml` file:

mapred.reduce.tasks

Specifies the number of reduce tasks for a job. Default is -1, which enables Hive to automatically determine the number of reducers for a job.

hive.exec.reducers.max

Determines the maximum number of reduce tasks that can be created for a job when you set the `mapred.reduce.tasks` property to -1.

hive.exec.reducers.bytes.per.reducer

Determines the number of reducers when you specify the size of data for each reducer. Default is 1 GB. For example, for an input size of 10 GB, 10 reducers are used.

For properties related to reduce tasks, default values work suitably for most mappings. However, for mappings that use a Data Processor transformation and a complex file writer, the values of these properties might require an increased number of reduce tasks.

hadoopEnv.properties

Configure the following property in the `hadoopEnv.properties` file in the `$INFA_HOME/services/shared/hadoop/<distr>/infaconf/hadoopEnv.properties` directory:

mapred.[reduce].child.java.opts

Because many map and reduce tasks run in parallel, Informatica recommends that you set the maximum heap size based on the system memory to reduce or avoid swapping tasks. For a parallel garbage collector, Informatica recommends that you set the `-XX:ParallelGCThreads` property to a smaller value such as 2 to reduce context switching overhead as the task tracker runs many map and reduce tasks.

For example, on a node with 32 GB RAM, with configurations that involve including 32 total slots along with map slots and reduce slots, you can specify the JVM maximum heap size value `-Xmx` to a maximum of 1 GB.

However, you cannot rule out swapping. Other operating system processes can compete for memory and the Data Transformation Manager (DTM) process that runs on the Hadoop node consumes memory outside the JVM heap. The default value of -Xmx512M is sufficient generally. However, Informatica recommends you to set the correct value after experimenting with different values to ensure swapping does not impact performance.

Tune the Mapping

You can tune the following parameters at the mapping level in the Developer Tool to optimize the performance:

- Ports precision
- Data movement mode
- Data type mapping

For more information, see the *Informatica Performance Tuning Guide*.

Ports Precision

Precision is the maximum number of significant digits for numeric data types, or maximum number of characters for string data types. For numeric data types, precision includes scale.

You can tune the precision in Model repository mappings.

When mappings contain ports with a larger precision than required, the mapping performance degrades. Informatica recommends that you set the precision judiciously for all source ports, transformation ports, and target ports. For instance, if a string port can handle data of a maximum of 200 characters, set the precision to 200. Do not set the precision to a high value such as 1000.

Data Movement Mode

When you run mappings on the Integration Service, the logical Data Transformation Manager (LDTM) component of the Integration Service determines whether to use the ASCII or Unicode data movement mode for mappings that read from a flat file or relational source. The LDTM determines the data movement mode based on the character set that the

mapping processes. When a mapping processes all ASCII data, the LDTM selects the ASCII mode. In ASCII mode, the Integration Service uses one byte to store each character, which can optimize mapping performance. In Unicode mode, the Integration Service uses two bytes for each character.

For more information about the data movement mode, see the "Data Integration Service Architecture" chapter in the *Informatica 10.1.1 Application Service Guide*.

Data Type Mapping

When the Integration Service reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Integration Service writes data to a target, it converts the transformation data types to the comparable native data types.

When you map source ports to transformation ports and then to target ports, avoid unnecessary data type conversions. For instance, do not map a port of the string data type to a port of the date data type. Ensure that you map ports to the same data type in all components of the mapping. Also, remove all unconnected ports from the mapping.

Use Hive Connection for Read and Write Operations

You can use a Hive connection to read data from and write data to Amazon S3.

When you run a mapping on the Blaze engine, a Hive connection reads and writes data to target files directly without creating the staging files. Writing data directly to Amazon S3 target improves the performance. For Amazon EMR distribution, Informatica recommends to use Hive connection in place of Amazon S3 connection for read and write operations. The Amazon S3 bucket and the Hadoop cluster must reside in the same region.

To use Hive connection for read and write operations on Amazon S3, perform the following steps:

- Create a Hive table on Amazon S3
- Configure core-site.xml

Create a Hive table on Amazon S3

Before you create a mapping to read or write data, create a Hive table for the Amazon S3 source and target. The following snippet shows the SQL command to create a sample table in Hive:

```
CREATE EXTERNAL TABLE lineitem (l_orderkey int, l_partkey int , l_suppkey int , l_linenumber int, l_quantity decimal(15,2), l_extendedprice decimal(15,2), l_discount decimal(15,2), l_tax decimal(15,2), l_returnflag string, l_linestatus string, l_shipdate string, l_commitdate string, l_receiptdate string, l_shipinstruct string, l_shipmode varchar(10), l_comment STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY "|" LOCATION 's3a://us-standard-bucket-test/S3_hive/';
```

The command creates a table named `lineitem` and the table points to the Amazon S3 location, `us-standard-bucket-test/S3_hive/`. Informatica recommends to use s3a URI scheme to connect to Amazon S3 through Hive connection.

Configure core-site xml

On each Hadoop node, update the core-site.xml file to add the following details to connect to Amazon S3:

- Filesystem
- S3 Endpoint
- Access Key ID
- Secret Access Key ID

The following snippet shows the sample `core-site.xml`:

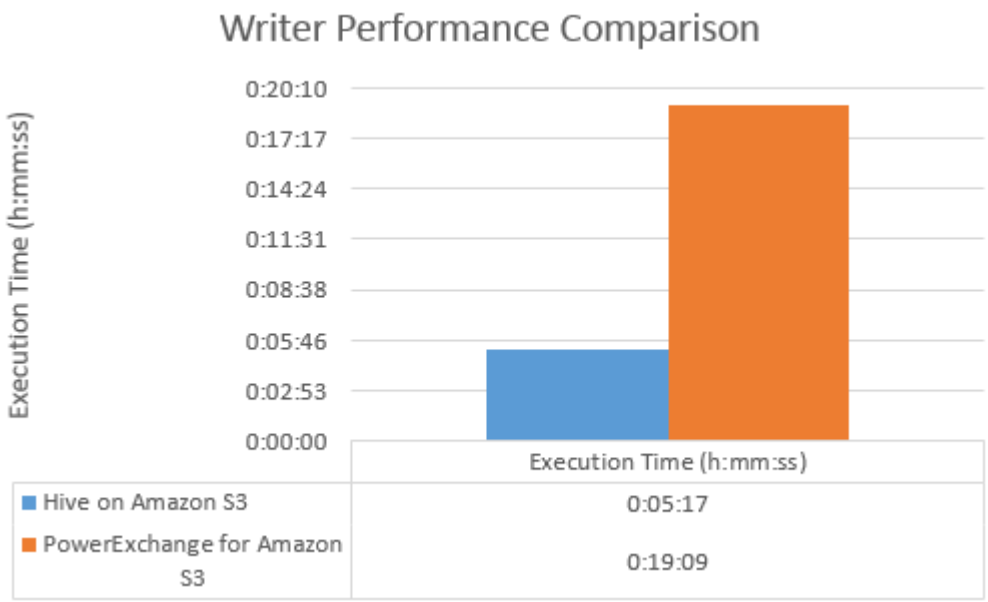
```
<property>
```

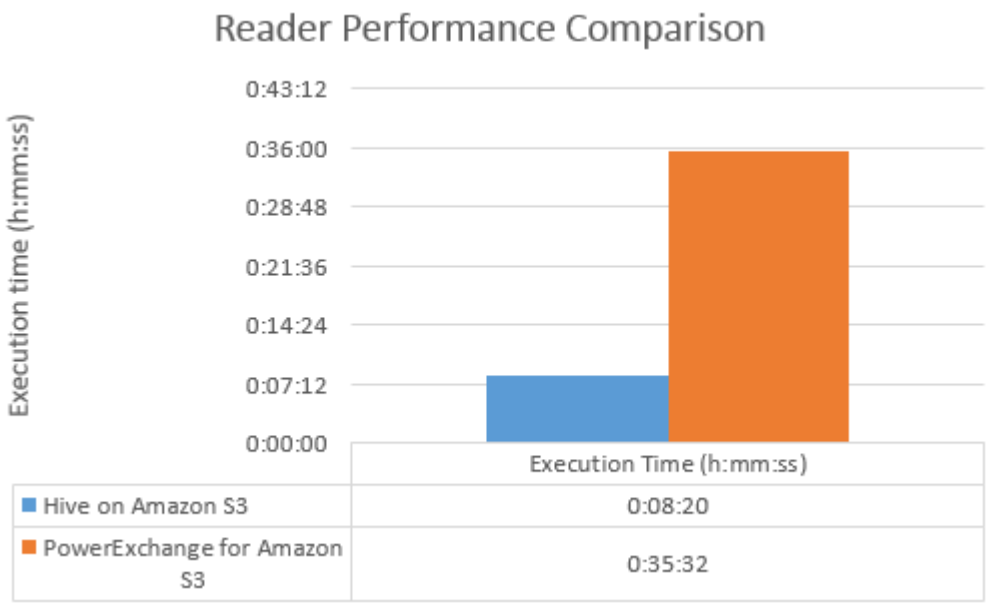
```
<name>fs.s3a.impl</name>
<value>com.amazon.ws.emr.hadoop.fs.EmrFileSystem</value>
</property>
<property>
<name>fs.s3a.endpoint</name>
<value>s3.amazonaws.com</value>
</property>
<property>
<name>fs.s3a.awsAccessKeyId</name>
<value>***************</value>
</property>
<property>
<name>fs.s3a.awsSecretAccessKey</name>
<value>*****************</value>
</property>
```

After creating the Hive table and updating the `core-site.xml` file, create a Hive connection. You can use the Hive connection in a mapping as a reader or writer.

Case Study

The following images show the Writer and Reader performance comparison between the PowerExchange for Amazon S3 connection and Hive connection on Amazon S3:





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